

IMO 2026

Complete Solutions

Kimi K3

Independently graded: **42/42**

Autonomous run, no human intervention.

Solutions as written by the model; all six problems.

github.com/deedy/imo-2026

IMO 2026 — Problem 1

Solution by Kimi K3

2026-07-17

Source: `results/kimi-k3/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Approaches tried

- **Lexicographic potential (product, #ones) + prime-wise gcd-of-valuations invariant** — successful; this is the proof below. See `approaches/invariant-valuations.md`.
- **Prime-wise reduction to a subtractive Euclidean game on exponents** — equivalent viewpoint, useful for discovering the invariant; see `approaches/prime-wise-euclidean.md`.
- **Dead ends** (recorded in `approaches/dead-ends.md`): the sum of the entries is *not* monotone (e.g. $(6, 10) \mapsto (2, 15)$, $16 \rightarrow 17$); the ordinary gcd of all entries is *not* the answer (e.g. $(4, 6) \mapsto \dots \mapsto M = 6 \neq 2$); an associative “star product” reformulation and induction on N were considered but not needed.

Current best

Complete solution. (a) Every play terminates: with P = product of all entries and z = number of entries equal to 1, a move with $g = \gcd(m, n)$ sends $P \mapsto P/g$; if $g \geq 2$ the product at least halves, if $g = 1$ the product is kept but z increases by 1 (and z never decreases), so there are at most $\lfloor \log_2 P_0 \rfloor + 2026$ moves. Moreover no move can produce two 1’s, so some entry > 1 always

survives, and the terminal position has **exactly one** entry $M > 1$. (b) For every prime p the quantity $G_p = \gcd(v_p(a_1), \dots, v_p(a_{2026}))$ is invariant, since a move changes the pair of valuations by $(\alpha, \beta) \mapsto (\min(\alpha, \beta), |\alpha - \beta|)$, which preserves gcd's. In the terminal position $G_p = v_p(M)$, whence the explicit choice-independent answer

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}$$

(the greatest common *exponential* divisor of the initial numbers). Verified exhaustively by computer for all small boards (`code/verify.py`).

Full proof

Let $N = 2026$ and let $a_1, \dots, a_N$ denote the integers on the board (so initially $a_i \geq 2$). A *move* chooses indices $i \neq j$ with $a_i, a_j > 1$ and replaces the pair $(a_i, a_j) = (m, n)$ by

$$g = \gcd(m, n) \quad \text{and} \quad \ell = \frac{\text{lcm}(m, n)}{\gcd(m, n)}.$$

Both g and ℓ are positive integers, because $g \mid \text{lcm}(m, n)$. Note also

$$g \cdot \ell = \text{lcm}(m, n) = \frac{mn}{g}. \quad (1)$$

Part (a): the process terminates with exactly one integer greater than 1

Step 1: every sequence of moves is finite. Let $P = a_1 a_2 \cdots a_N \in \mathbb{Z}_{>0}$ be the product of all entries and let $z = \#\{i : a_i = 1\}$. Consider a move on (m, n) and put $g = \gcd(m, n)$.

Effect on P . By (1), the product of the two affected entries changes from mn to mn/g ; all other entries are unchanged. Hence P becomes P/g .

Monotonicity of z . A move never destroys an entry equal to 1: only entries > 1 may be chosen, so the two replaced entries exceed 1, and the two new entries are ≥ 1 . Hence z is non-decreasing throughout the process; also $0 \leq z \leq N$.

Type A: $g \geq 2$. Then P is multiplied by $1/g \leq \frac{1}{2}$.

Type B: $g = 1$. Then $\ell = mn \geq 2 \cdot 2 = 4 > 1$, so the pair (m, n) is replaced by $(1, mn)$: the product P is unchanged and z increases by exactly 1.

Now bound the number of moves of each type. Since z is non-decreasing, takes values in $\{0, 1, \dots, N\}$, and increases by 1 at every type B move, the whole play contains **at most N type B moves**. Since P is always a positive integer and each type A move at least halves it, if the play contained t type A moves then $1 \leq P_{\text{final}} \leq P_0/2^t$, so $t \leq \lfloor \log_2 P_0 \rfloor$, where P_0 is the initial product. Therefore every play has at most $\lfloor \log_2 P_0 \rfloor + N$ moves; in particular every play is finite, regardless of Confucius's choices. ■

Step 2: at least one entry greater than 1 always remains. We prove by induction on the number of moves that after any number of moves, some entry exceeds 1. This holds initially since all N entries exceed 1. Suppose it holds before a move on (m, n) . If $g = \gcd(m, n) \geq 2$, then the new entry g itself exceeds 1. If $g = 1$, then the new entry $\ell = mn \geq 4$ exceeds 1. All entries not involved in the move are unchanged, so after the move some entry still exceeds 1. ■

Conclusion of (a). Confucius continues while a move is possible, i.e. while there are two entries > 1 at different places. By Step 1 he stops after finitely many moves; in the final position **at most one** entry exceeds 1, and by Step 2 **at least one** entry exceeds 1. Hence exactly one entry $M > 1$ remains, and the other $N - 1$ entries all equal 1. ■

Part (b): the final value M is independent of the choices

For a prime p and an integer $x \geq 1$, write $v_p(x)$ for the exponent of p in the factorisation of x (so $v_p(1) = 0$). For each prime p define

$$G_p = \gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_N)) \in \mathbb{Z}_{\geq 0},$$

the greatest common divisor of a tuple of nonnegative integers, with the convention $\gcd(0, \dots, 0) = 0$.

Step 3: G_p is invariant under moves. A move changes only two of the N entries, hence only two of the N valuations, namely

$$\alpha = v_p(m), \quad \beta = v_p(n) \longrightarrow \alpha' = v_p(g) = \min(\alpha, \beta), \quad \beta' = v_p(\ell) = \max(\alpha, \beta) - \min(\alpha, \beta) = |\alpha - \beta|.$$

Claim: for every positive integer t ,

$$t \mid \alpha \text{ and } t \mid \beta \iff t \mid \alpha' \text{ and } t \mid \beta'.$$

Indeed, we may assume $\alpha \leq \beta$, so $\alpha' = \alpha$ and $\beta' = \beta - \alpha$. If $t \mid \alpha$ and $t \mid \beta$, then $t \mid \beta - \alpha$. Conversely, if $t \mid \alpha$ and $t \mid \beta - \alpha$, then $t \mid \alpha + (\beta - \alpha) = \beta$. This proves the claim.

Consequently the set of positive common divisors of the whole N -tuple of valuations is unchanged by the move. If the tuple is not identically zero, its gcd is the largest element of this set, hence unchanged. If the tuple is identically zero, then in particular $\alpha = \beta = 0$, so $\alpha' = \beta' = 0$ and the tuple remains identically zero; the gcd equals 0 before and after. In all cases G_p is invariant. ■

Step 4: computing M . By part (a), every play ends in the position $(1, 1, \dots, 1, M)$ (up to the order of the places). In this position the tuple of p -adic valuations is $(0, \dots, 0, v_p(M))$, whose greatest common divisor equals $v_p(M)$, since $\gcd(x, 0) = x$. By the invariance of Step 3 applied to the initial numbers $a_1, \dots, a_N$,

$$v_p(M) = \gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_N)) \quad \text{for every prime } p.$$

A positive integer is uniquely determined by its exponents $v_p(\cdot)$, and only primes dividing some initial a_i contribute; hence

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_N))}$$

which depends only on the initial integers, not on Confucius's choices. ■

Remarks (consistency checks). - If all initial numbers are equal to c , then $M = c$: each move is $(c, c) \mapsto (c, 1)$. - If the initial numbers are pairwise coprime, each valuation tuple contains at most one nonzero entry, and $\gcd(0, \dots, 0, e) = e$, so $M = a_1 a_2 \cdots a_N$: coprime moves $(m, n) \mapsto (1, mn)$ merely merge everything into one product. - For two initial numbers (x, y) one gets $M = \prod_p p^{\gcd(v_p(x), v_p(y))}$; e.g. $(4, 6) \mapsto (2, 6) \mapsto (2, 3) \mapsto (1, 6)$ gives $M = 6 = 2^{\gcd(2, 1)} 3^{\gcd(0, 1)}$. -

In the language of exponential divisibility, M is the greatest common exponential divisor of the initial numbers: the unique maximal d with $v_p(d) \mid v_p(a_i)$ for all p and all i .

Computer verification (`code/verify.py`): exhaustive search over all plays for all multisets of lengths 2–6 with small values confirms that every play ends with exactly one entry > 1 and that the terminal value coincides with the boxed formula; 500 random playouts with $N \leq 12$ and 2000 single-move invariant checks confirm the same. All tests passed.

IMO 2026 — Problem 2

Solution by Kimi K3

2026-07-17

Source: `results/kimi-k3/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Approaches tried

- **Numerical exploration** (`code/explore.py`): confirmed $OM = ON$ across many triangles and parameter choices; confirmed the configuration exists and is (generically) unique for each φ .
- **Power-of-a-point reformulation:** $OM = ON \iff \text{pow}(M) = \text{pow}(N) \iff \text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}$ (median/Apollonius identity). This is the route that worked.
- **Angle chasing:** found $\cot \angle BAK = \cot \varphi + 2 \cot \gamma$, $\cot \angle CAL = \cot \varphi + 2 \cot \beta$; found $\angle MKC = \angle NLB = 180^\circ - A - \varphi$ (automatic, not the crux).
- **Law of sines in four triangles** (BMK , CNL , BKC , BCL) $\rightarrow$ two quadratic constraints ConsP , ConsQ on $p = BK$, $q = CL$. **This is the key structural fact.**
- **Coordinate computation of powers** with A at origin (complex notation $K = c - pe^{-i\varphi}$, $L = e^{iA}(b - qe^{i\varphi})$) $\rightarrow$ explicit formulas for $\text{pow}(B)$, $\text{pow}(C)$; discovered they simplify to closed forms using ConsP , ConsQ .
- **Symbolic verification** (`code/verify_identity.py`, `code/verify_power_formulas.py`): the two crucial polynomial identities are exactly true (difference simplifies to 0); the elimination R1 R3 ConsP verified exactly.

Current best

Complete rigorous proof. The crux: the given angle conditions force $p = BK$ and $q = CL$ to satisfy the quadratics $2\sin(A + \varphi)p^2 - (2\sin A + \sin(A + 2\varphi))cp + c^2\sin(A + \varphi) - bc\sin\varphi = 0$ (and the analogous one for q with b), which in turn force $\text{pow}(B) = \frac{c^2\sin(A + \varphi) - bc\sin\varphi}{2\sin(A + \varphi)}$ and $\text{pow}(C) = \frac{b^2\sin(A + \varphi) - bc\sin\varphi}{2\sin(A + \varphi)}$, so $\text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}$, equivalent to $OM = ON$.

Full proof**Notation**

Let A, B, C also denote the angles of $\triangle ABC$, and $a = BC$, $b = CA$, $c = AB$. Put

$$\varphi := \angle KBA = \angle ACL, \quad \beta := \angle LBK = \angle LNC, \quad \gamma := \angle LCK = \angle BMK,$$

and $p := BK$, $q := CL$. Since K lies inside $\angle LBA$, $\angle ABL = \beta + \varphi$; since L lies inside $\angle ACK$, $\angle ACK = \gamma + \varphi$. All angles appearing below are positive because $K \in \triangle BMC$ and $L \in \triangle BNC$ are interior points (so rays BK, BL lie inside $\angle ABC$ and rays CK, CL inside $\angle ACB$); in particular $0 < \varphi < B$, $\beta + \varphi < B$, $\gamma + \varphi < C$, whence $A + \varphi < A + B < \pi$ and $\sin(A + \varphi) > 0$.

Step 1: Reduction to powers of B and C

Let R be the radius of the circumcircle $\omega = (AKL)$ and write $\text{pow}(\cdot)$ for power w.r.t. ω . Since M is the midpoint of AB , Apollonius's median formula in $\triangle OAB$ gives

$$OM^2 = \frac{2OA^2 + 2OB^2 - AB^2}{4} = \frac{2R^2 + 2OB^2 - c^2}{4},$$

so $\text{pow}(M) = OM^2 - R^2 = \frac{\text{pow}(B)}{2} - \frac{c^2}{4}$. Similarly $\text{pow}(N) = \frac{\text{pow}(C)}{2} - \frac{b^2}{4}$. Therefore

$$OM = ON \iff \text{pow}(M) = \text{pow}(N) \iff \text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}. \quad (1)$$

Step 2: Four law-of-sines relations

Triangle BMK: angles φ at B (as $M \in BA$), γ at M , $\pi - \varphi - \gamma$ at K ; with $BM = c/2$:

$$p = BK = \frac{c \sin \gamma}{2 \sin(\varphi + \gamma)}. \quad (R1)$$

Triangle CNL: angles φ at C (as $N \in CA$), β at N , $\pi - \varphi - \beta$ at L ; with $CN = b/2$:

$$q = CL = \frac{b \sin \beta}{2 \sin(\varphi + \beta)}. \quad (R2)$$

Triangle BKC: $\angle KBC = B - \varphi$, $\angle KCB = C - (\gamma + \varphi)$ (using $\angle ACK = \gamma + \varphi$), $\angle BKC = A + \gamma + 2\varphi$:

$$p = BK = \frac{a \sin(C - \varphi - \gamma)}{\sin(A + \gamma + 2\varphi)}. \quad (R3)$$

Triangle BCL: $\angle LBC = B - (\beta + \varphi)$ (using $\angle ABL = \beta + \varphi$), $\angle BCL = C - \varphi$, $\angle BLC = A + \beta + 2\varphi$:

$$q = CL = \frac{a \sin(B - \varphi - \beta)}{\sin(A + \beta + 2\varphi)}. \quad (R4)$$

Step 3: The quadratic constraints (elimination of β, γ)

Auxiliary identity: for any x , $c \sin(A + x) = b \sin x + a \sin(C - x)$. Indeed with $a = 2R' \sin A$, $b = 2R' \sin B$, $c = 2R' \sin C$ this is equivalent to $\sin C \sin(A + x) = \sin B \sin x + \sin A \sin(C - x)$, and using $\sin B = \sin(A + C)$ the RHS equals $\sin C(\sin x \cos A + \sin A \cos x) = \sin C \sin(A + x)$.

Combining (R3) with the identity at $x = \gamma + \varphi$:

$$p \sin(A + \gamma + 2\varphi) = c \sin(A + \gamma + \varphi) - b \sin(\gamma + \varphi). \quad (\text{EqC})$$

Expanding (R1) and (EqC) with the addition formulas:

$$2p \sin \varphi \cos \gamma = (c - 2p \cos \varphi) \sin \gamma, \quad (i)$$

$$[c \cos(A + \varphi) - p \cos(A + 2\varphi) - b \cos \varphi] \sin \gamma = [p \sin(A + 2\varphi) - c \sin(A + \varphi) + b \sin \varphi] \cos \gamma. \quad (ii)$$

Eliminating $\sin \gamma, \cos \gamma$ from (i),(ii) (multiply (i) by the bracket in (ii)'s LHS and (ii) by $2p \sin \varphi$, subtract — legitimate since the resulting identity $(c - 2p \cos \varphi)(\dots) - (2p \sin \varphi)(\dots) = 0$ follows from (i) (ii)):

$$(c - 2p \cos \varphi)[p \sin(A + 2\varphi) - c \sin(A + \varphi) + b \sin \varphi] = 2p \sin \varphi[c \cos(A + \varphi) - p \cos(A + 2\varphi) - b \cos \varphi].$$

Expand. The terms $-2pb \sin \varphi \cos \varphi$ cancel on both sides. Using $2 \sin \varphi \cos(A + \varphi) = \sin(A + 2\varphi) - \sin A$, $2 \cos \varphi \sin(A + \varphi) = \sin(A + 2\varphi) + \sin A$, and $\cos \varphi \sin(A + 2\varphi) - \sin \varphi \cos(A + 2\varphi) = \sin(A + \varphi)$, everything simplifies to:

$$\boxed{2 \sin(A + \varphi) p^2 - (2 \sin A + \sin(A + 2\varphi))cp + c^2 \sin(A + \varphi) - bc \sin \varphi = 0} \quad (\text{ConsP})$$

The symmetric computation with (R2),(R4) (swap $b \leftrightarrow c$, $p \leftrightarrow q$, $\gamma \leftrightarrow \beta$) gives:

$$\boxed{2 \sin(A + \varphi) q^2 - (2 \sin A + \sin(A + 2\varphi))bq + b^2 \sin(A + \varphi) - bc \sin \varphi = 0} \quad (\text{ConsQ})$$

Step 4: Powers of B and C in coordinates

Place $A = (0, 0)$, $B = (c, 0)$, $C = (b \cos A, b \sin A)$. From $\angle KBA = \varphi$, $BK = p$ and from $\angle ACL = \varphi$, $CL = q$:

$$K = (c - p \cos \varphi, p \sin \varphi), \quad L = (b \cos A - q \cos(A + \varphi), b \sin A - q \sin(A + \varphi)).$$

The circle ω passes through the origin, so it has equation $x^2 + y^2 + ux + vy = 0$; the power of a point P equals $g(P) = x_P^2 + y_P^2 + ux_P + vy_P$. The conditions $g(K) = g(L) = 0$ form a 2×2 linear system in u, v with determinant

$$\Delta = K_x L_y - K_y L_x = bc \sin A - (bp + cq) \sin(A + \varphi) + pq \sin(A + 2\varphi) = 2[AKL] \neq 0$$

(A, K, L are not collinear since ω is the circumcircle of a genuine triangle). Solving:

$$u = \frac{|L|^2 K_y - |K|^2 L_y}{\Delta}, \quad v = \frac{|K|^2 L_x - |L|^2 K_x}{\Delta},$$

where $|K|^2 = c^2 - 2cp \cos \varphi + p^2$ and $|L|^2 = b^2 - 2bq \cos \varphi + q^2$. Therefore

$$\text{pow}(B) = c^2 + cu = c^2 + \frac{c[|L|^2 p \sin \varphi - |K|^2 (b \sin A - q \sin(A + \varphi))]}{\Delta}, \quad (2)$$

and, using $L_x \sin A - L_y \cos A = q \sin \varphi$ and $K_y \cos A - K_x \sin A = p \sin(A + \varphi) - c \sin A$,

$$\text{pow}(C) = b^2 + ub \cos A + vb \sin A = b^2 + \frac{b[|K|^2 q \sin \varphi + |L|^2 (p \sin(A + \varphi) - c \sin A)]}{\Delta}. \quad (3)$$

Step 5: Key algebraic identities

Write $s_1 = \sin(A + \varphi)$, $s_\varphi = \sin \varphi$, $s_A = \sin A$. The following two identities hold, where each side is a polynomial in p, q whose coefficients are trigonometric polynomials in A, φ :

$$2s_1 \Delta \left(\text{pow}(B) - \frac{c^2 s_1 - bcs_\varphi}{2s_1} \right) = c(s_1 q - bs_A) \cdot \text{ConsP} + cs_\varphi p \cdot \text{ConsQ}, \quad (4)$$

$$2s_1 \Delta \left(\text{pow}(C) - \frac{b^2 s_1 - bcs_\varphi}{2s_1} \right) = bs_\varphi q \cdot \text{ConsP} + b(s_1 p - cs_A) \cdot \text{ConsQ}, \quad (5)$$

where $\text{pow}(B), \text{pow}(C)$ are substituted from (2),(3) and $\text{ConsP}, \text{ConsQ}$ from Step 3.

Verification of (4). Multiplying (2) out, the LHS of (4) equals

$$\Delta(c^2 s_1 + bcs_\varphi) + 2s_1 c \left[|L|^2 ps_\varphi - |K|^2 (bs_A - qs_1) \right].$$

Both this and the RHS of (4) are polynomials in p, q involving only the monomials $p^2 q, pq^2, p^2, q^2, pq, p, q, 1$. Comparing coefficients monomial by monomial, the two sides agree immediately in every monomial except possibly pq ; the difference of the two pq -coefficients is $c \cdot E(A, \varphi)$ with

$$E = b[3s_A s_\varphi + 4s_\varphi^2 \cos(A + \varphi) - \cos A \cos \varphi + \cos(A + 3\varphi)] + 4c[s_A s_1 + \cos A \cos(A + \varphi) - \cos \varphi].$$

The second bracket is $\cos((A + \varphi) - A) - \cos \varphi = 0$. For the first, write $\cos(A + 3\varphi) = \cos(A + \varphi) \cos 2\varphi - \sin(A + \varphi) \sin 2\varphi$; then

$$\begin{aligned} s_\varphi^{-1} \cdot (\text{first bracket}) &= 3s_A s_\varphi + 4s_\varphi^2 \cos(A + \varphi) - \cos A \cos \varphi + \cos(A + \varphi)(1 - 2s_\varphi^2) - 2s_1 s_\varphi \cos \varphi \\ &= 3s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) + 2s_\varphi [s_\varphi \cos(A + \varphi) - s_1 \cos \varphi] \\ &= 3s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) + 2s_\varphi \sin(\varphi - (A + \varphi)) \\ &= s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) = -\cos(A + \varphi) + \cos(A + \varphi) = 0. \end{aligned}$$

Hence $E = 0$ and (4) holds. (All steps are direct expansions using only the addition formulas; the computation was also verified symbolically in `code/verify_power_formulas.py`.) Identity (5) is the same computation with the roles $b \leftrightarrow c$, $p \leftrightarrow q$ interchanged.

Step 6: Conclusion

Since $\text{ConsP} = \text{ConsQ} = 0$ (Step 3) and $s_1 \Delta \neq 0$, identities (4),(5) give

$$\text{pow}(B) = \frac{c^2 \sin(A + \varphi) - bc \sin \varphi}{2 \sin(A + \varphi)}, \quad \text{pow}(C) = \frac{b^2 \sin(A + \varphi) - bc \sin \varphi}{2 \sin(A + \varphi)}.$$

Subtracting:

$$\text{pow}(B) - \text{pow}(C) = \frac{(c^2 - b^2) \sin(A + \varphi)}{2 \sin(A + \varphi)} = \frac{c^2 - b^2}{2}.$$

By (1) this is equivalent to $OM = ON$. ■

IMO 2026 — Problem 3

Solution by Kimi K3

2026-07-19

Source: `results/kimi-k3/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Solution document

Status

solved

Approaches tried

- Modeling: claiming game with “pick any piece” — optimal play = greedy (proved below, brute-force verified). Game reduces to: LB picks a composition of 1 into $\leq n + 1$ pieces, XY subdivides with $\leq n$ cuts; payoff D = alternating sum; LB share = $(1 + D)/2$.
- Grid computation: $c_1 = 2/3, c_2 = 4/7, c_3 = 8/15$ conjecture $c_n = \frac{2^n}{2^{n+1}-1}$, LB plays geometric pieces $\frac{2^{i-1}}{2^{n+1}-1}$. (Confirmed as the answer.)
- Odd-set identity: $D(M) = |\{t : N(t) \text{ odd}\}|$ (layer cake). Basis of the matching formula.
- **Matching formula** (proved below; verified `code/tests2.py`, 300 tests): $D(M) = \min_{\mu} (\sum_{\{x,y\} \in \mu} |x - y| + \sum_{z \text{ unmatched}} z)$.
- **Part A** (LB guarantee): super-increasing family (margin δ , m pieces), subdivided with $\leq m - 1$ cuts $D \geq \delta$. Proof: matching formula + loop-augmented component graph + counting ($e \leq m - 1$ tree component exists) + tree inequality ($|\sum_{i \in C} \epsilon_i a_i| \leq \text{cost}(C)$) + margin. Verified 1500 random trials. (Fix vs. earlier sketch: self-pairs count as *loops*, making their component cyclic; with this, the tree inequality is per-component correct.)

- NM (“some same-parent merge never increases D ”) — FALSE in general ($\{1,3\}$, cut $3 \rightarrow 1.5+1.5$: D drops $2 \rightarrow 1$, only merge raises it). Induction-via-merging abandoned; matching route used instead.
- **Part B** (XY counterstrategy): parity view of cuts: bisect x (=remove x), cancel $x > y$ (=remove x, y , add $x - y$), 1 cut each. Chains = nested abs values on a subset. Chain theorem: any m positive reals with sum S have a nonempty subset **and an ordering of it** with chain value $\leq S/(2^m - 1)$; proved by pigeonhole on 2^m subset sums + **greedy-scheduling realizability lemma** (adjacent gaps are chain-realizable). Verified $n=1..4$ (2000 random each) + adversarial local search (extremum = geometric family, value exactly $T_n S$); AGR checked 180 families, 0 failures.
- Dominance “min-subdiv = min-halve/cancel”: false on coarse grids (grid artifacts only); irrelevant — for Part B we only need halve/cancel sequences, which are valid subdivisions.
- Independent-review audit (final session): Lemma 1.1 turn-parity fix (second player secures $\text{even}(M'')$, not $\text{odd}(M'')$) re-verified line-by-line for all j and by exact minimax on 304 multisets (reviewer example $\{3, 2, 1\}$: second player gets exactly 2 = $\text{even}(M)$); Theorem 4.2 ordering fix confirmed — the theorem produces the ordering and the strategy processes it. Four presentation patches applied (empty-multiset base case in Lemma 1.1; isolated-vertex components in Theorem 3.1; precise induction step in the sharpness remark; running-piece validity in XY’s strategy).

Current best

Answer: $c = \frac{2^n}{2^{n+1}-1}$. Proved in full below. Lower bound (LB guarantee): LB cuts into geometric pieces $\frac{2^{i-1}}{2^{n+1}-1}$ ($i = 1, \dots, n+1$); any refinement by $\leq n$ further cuts has alternating sum $D \geq \frac{1}{2^{n+1}-1}$ (Theorem 3.1). Upper bound (XY counterstrategy): for any LB composition, XY either bisects all pieces (if fewer than $n+1$) forcing $D = 0$, or uses a sub- T_n chain on some subset, suitably ordered (Theorem 4.2, via pigeonhole + realizability), to force $D \leq \frac{1}{2^{n+1}-1}$. Hence LB gets exactly $\frac{1}{2}(1 + \frac{1}{2^{n+1}-1}) = \frac{2^n}{2^{n+1}-1}$ under optimal play.

Full proof

Setup and notation. After all marks are made, the stick is cut into a finite multiset M of positive piece lengths summing to 1. For a multiset $M = \{p_1 \geq p_2 \geq \dots \geq p_m\}$ sorted in decreasing order, define the *alternating sum*

$$D(M) = p_1 - p_2 + p_3 - p_4 + \dots = \sum_{i=1}^m (-1)^{i+1} p_i,$$

and write $\text{odd}(M) = \sum_{i \text{ odd}} p_i$, $\text{even}(M) = \sum_{i \text{ even}} p_i$ for the sums of the odd- and even-indexed pieces. Note $\text{odd} + \text{even} = \sum M$ and $\text{odd} - \text{even} = D(M) \geq 0$ (since $D = (p_1 - p_2) + (p_3 - p_4) + \dots \geq 0$).

1. The claiming game

Lemma 1.1 (value of the claiming game). *In the claiming game played on a multiset M of positive reals — two players alternately take any remaining piece, each maximising his own total — the first player can guarantee a total of at least $\text{odd}(M)$, and the second player can guarantee a total of at least $\text{even}(M)$. Since $\text{odd}(M) + \text{even}(M) = \sum M$, under optimal play the first player*

obtains exactly $\text{odd}(M) = \frac{\sum M + D(M)}{2}$ and the second player exactly $\text{even}(M)$. In particular, in the game of the problem Liu Bang, who moves first, obtains exactly $\text{odd}(M)$.

Proof. Induction on $m = |M|$, with the convention $\text{odd}(\emptyset) = \text{even}(\emptyset) = 0$ (needed when $m = 2$, since the inductive step passes to a multiset of size $m - 2$). If $m \leq 1$ both claims are immediate: for $m = 1$ the first player takes $p_1 = \text{odd}(M)$ and the second player gets $0 = \text{even}(M)$. Now let $m \geq 2$ and sort $M = \{p_1 \geq \dots \geq p_m\}$.

First player's guarantee. The first player takes p_1 . Suppose the second player then takes p_j ($j \geq 2$). On the remainder $M' = M \setminus \{p_1, p_j\}$ the first player moves first, so by induction he secures at least $\text{odd}(M')$ from M' , for a total of $p_1 + \text{odd}(M')$. The sorted order of M' is $p_2, \dots, p_{j-1}, p_{j+1}, \dots, p_m$; its odd-indexed elements are the p_i with i even, $2 \leq i \leq j - 1$, together with the p_i with i odd, $i \geq j + 1$. Therefore

$$(p_1 + \text{odd}(M')) - \text{odd}(M) = \sum_{\substack{i \text{ even} \\ 2 \leq i \leq j-1}} p_i - \sum_{\substack{i \text{ odd} \\ 3 \leq i \leq j}} p_i = (p_2 - p_3) + (p_4 - p_5) + \dots \geq 0,$$

the right-hand side being a sum of consecutive differences, ending with $(p_{j-2} - p_{j-1})$ if j is even and with $(p_{j-1} - p_j)$ if j is odd (empty if $j = 2$). So the first player secures at least $\text{odd}(M)$ no matter what the second player does.

Second player's guarantee. Suppose the first player takes p_j . The second player takes a largest remaining piece ℓ ($\ell = p_1$ if $j \geq 2$, $\ell = p_2$ if $j = 1$). On the remainder $M'' = M \setminus \{p_j, \ell\}$ it is the first player's turn, so the second player of the original game is the **second** player on M'' ; by induction he secures at least $\text{even}(M'')$ from M'' , for a total of at least $\ell + \text{even}(M'')$.

If $j = 1$: $M'' = \{p_3, \dots, p_m\}$, whose even-indexed elements are $p_4, p_6, \dots$; hence

$$\ell + \text{even}(M'') = p_2 + (p_4 + p_6 + \dots) = \text{even}(M).$$

If $j \geq 2$: $M'' = M \setminus \{p_1, p_j\}$ has sorted order $p_2, \dots, p_{j-1}, p_{j+1}, \dots, p_m$; its even-indexed elements are the p_i with i odd, $3 \leq i \leq j - 1$, together with the p_i with i even, $i \geq j + 1$. Hence

$$(\ell + \text{even}(M'')) - \text{even}(M) = p_1 + \sum_{\substack{i \text{ odd} \\ 3 \leq i \leq j-1}} p_i - \sum_{\substack{i \text{ even} \\ 2 \leq i \leq j}} p_i = (p_1 - p_2) + (p_3 - p_4) + \dots \geq 0,$$

a sum of consecutive differences, ending with $(p_{j-1} - p_j)$ if j is even and with $(p_{j-2} - p_{j-1})$ if j is odd (it reduces to $p_1 - p_2$ if $j = 2$). So the second player secures at least $\text{even}(M)$ no matter what the first player does. ■

Since $\text{odd}(M) + \text{even}(M) = \sum M = 1$, LB's share under optimal play equals $\text{odd}(M) = \frac{1 + D(M)}{2}$. Consequently the whole game is equivalent to the following one-dimensional optimization:

Reduction. LB chooses a composition of 1 into $m \leq n + 1$ positive parts (his $m - 1 \leq n$ marks). XY then refines this composition with at most n further cuts (each of XY's marks lies in the interior of one current piece and splits it in two; conversely every such refinement is achievable, with all marks distinct, since cut points are interior to pieces). LB's resulting share is $\frac{1 + D(M)}{2}$, where M is the final multiset of pieces.

We therefore must compute $D^* = \max_{\text{LB}} \min_{\text{XY}} D(M)$; the answer will be $c = \frac{1 + D^*}{2}$.

2. Two representations of D

Lemma 2.1 (odd-set identity). *For a finite multiset M of positive reals and $t \geq 0$ let $N(t) = \#\{x \in M : x > t\}$. Then*

$$D(M) = |\{t \geq 0 : N(t) \text{ is odd}\}|,$$

the Lebesgue measure of the “odd set”.

Proof. Sort $M = \{p_1 \geq \dots \geq p_m\}$ and put $p_{m+1} = 0$. For $t \in [p_{i+1}, p_i]$ we have $N(t) = i$ (ties give empty intervals, which do not affect the measure). Hence $|\{t : N(t) \text{ odd}\}| = \sum_{i \text{ odd}} (p_i - p_{i+1}) = D(M)$. ■

A *matching* μ of M is a collection of disjoint unordered pairs $\{x, y\}$ of elements of M (elements may be left unpaired); its *cost* is

$$\text{cost}(\mu) = \sum_{\{x, y\} \in \mu} |x - y| + \sum_{z \text{ unmatched}} z.$$

Lemma 2.2 (matching formula). $D(M) = \min_{\mu} \text{cost}(\mu)$, the minimum over all matchings of M .

Proof. For $t \geq 0$ and a matching μ let $c_{\mu}(t)$ be the number of pairs of μ with exactly one element $> t$, plus the number of unmatched elements $> t$. Since $|x - y| = \int_0^{\infty} \mathbf{1}[\text{exactly one of } x, y > t] dt$ and $z = \int_0^{\infty} \mathbf{1}[z > t] dt$, we have $\text{cost}(\mu) = \int_0^{\infty} c_{\mu}(t) dt$. If $b(t)$ denotes the number of pairs with both elements $> t$, then $N(t) = 2b(t) + c_{\mu}(t)$, so $c_{\mu}(t) \equiv N(t) \pmod{2}$ and $c_{\mu}(t) \geq 0$; in particular $c_{\mu}(t) \geq \mathbf{1}[N(t) \text{ odd}]$ for every t . Integrating and using Lemma 2.1 gives $\text{cost}(\mu) \geq D(M)$ for every μ . Equality is attained by the *adjacent* matching: sorting $M = \{p_1 \geq \dots \geq p_m\}$ and pairing $(p_1, p_2), (p_3, p_4), \dots$ gives $\text{cost} \sum_{i \text{ odd}} (p_i - p_{i+1}) = D(M)$. ■

Corollary 2.3 (parity cancellation). *If M contains two equal elements x, x , then $D(M) = D(M \setminus \{x, x\})$. Consequently, if every value in M occurs with even multiplicity, then $D(M) = 0$.*

Proof. In the decreasing ordering of M the two copies of x occupy two adjacent positions $j, j+1$ and contribute $(-1)^{j+1}x + (-1)^{j+2}x = 0$; removing them shifts all later elements by two positions, preserving the sign of each term. ■

3. Part A: the geometric split (lower bound)

Put $T_n = \frac{1}{2^{n+1} - 1}$. LB marks n points dividing the stick into $n+1$ pieces of lengths

$$a_i = 2^{i-1}T_n \quad (i = 1, \dots, n+1), \quad \sum_{i=1}^{n+1} a_i = (2^{n+1} - 1)T_n = 1.$$

Definition. Positive reals $a_1 \leq a_2 \leq \dots \leq a_m$ are *super-increasing with margin* $\delta > 0$ if $a_1 \geq \delta$ and $a_i \geq \delta + \sum_{j < i} a_j$ for every $i \geq 2$.

The family above is super-increasing with margin $\delta = T_n$: indeed $a_1 = T_n$ and $\sum_{j < i} a_j = (2^{i-1} - 1)T_n = a_i - T_n$, so $a_i = \delta + \sum_{j < i} a_j$.

Theorem 3.1. *Let $a_1 \leq \dots \leq a_m$ be super-increasing with margin δ , and let M be obtained from these pieces by at most $m-1$ cuts (each cut splits one current piece into two positive pieces). Then $D(M) \geq \delta$.*

Proof. By Lemma 2.2 it suffices to show that **every** matching μ of M has $\text{cost}(\mu) \geq \delta$. Fix μ . Each element of M is a fragment of exactly one piece a_i . Build a multigraph G on the vertex set $\{1, \dots, m\}$: - for each pair $\{x, y\} \in \mu$ whose fragments come from two **different** pieces $i \neq j$, add an edge $\{i, j\}$; - for each pair $\{x, y\} \in \mu$ whose two fragments come from the **same** piece i , add a **loop** at i .

Unmatched fragments create no edges. Write e for the total number of edges (loops included) and u for the number of unmatched fragments. Counting fragments in two ways, with $c \leq m - 1$ the number of cuts:

$$2e + u = \#\{\text{fragments}\} = m + c \leq 2m - 1,$$

so $e \leq m - 1$ (as e is an integer and $u \geq 0$). Let C range over the connected components of G — **including isolated vertices** (a piece all of whose fragments are unmatched is such a component, with $e_C = 0$, $v_C = 1$); connectivity ignores loops, but loops still count as edges of their component. Let e_C, v_C be the numbers of edges and vertices of C ; then $\sum_C v_C = m$ and $\sum_C e_C = e$. If $e_C \geq v_C$ held for **every** component, summing would give $e \geq m$, contradicting $e \leq m - 1$. Hence some component C has $e_C \leq v_C - 1$. A connected multigraph on v_C vertices has at least $v_C - 1$ edges, with equality only if it is a tree (a loop or a parallel edge would force $e_C \geq v_C$); so C is a tree. In particular C has **no loops**, i.e. no self-pair of fragments occurs inside C : every pair of μ within C is a cross-pair between two different pieces of C , and these cross-pairs are exactly the $v_C - 1$ edges of the tree.

Tree inequality. A tree is bipartite: choose signs $\epsilon_i \in \{\pm 1\}$ for $i \in C$ so that $\epsilon_i = -\epsilon_j$ whenever $\{i, j\}$ is an edge. We claim

$$\left| \sum_{i \in C} \epsilon_i a_i \right| \leq \text{cost}(C), \quad \text{cost}(C) := \sum_{\substack{\{x, y\} \in \mu \\ \text{within } C}} |x - y| + \sum_{\substack{z \text{ unmatched} \\ \text{in } C}} z.$$

Indeed, expand each a_i as the sum of its fragments and group the signed sum $\sum_{i \in C} \epsilon_i a_i$ according to μ . A cross-pair $\{x, y\}$ with x from i and y from j contributes $\epsilon_i x + \epsilon_j y = \epsilon_i(x - y)$, of absolute value $|x - y|$; an unmatched fragment z of piece i contributes $\epsilon_i z$, of absolute value z . By the triangle inequality $\left| \sum_{i \in C} \epsilon_i a_i \right| \leq \sum_{\text{cross-pairs}} |x - y| + \sum_{\text{unmatched}} z = \text{cost}(C)$.

Margin. Let k be the largest index in C . Then, using the reverse triangle inequality and the margin property,

$$\left| \sum_{i \in C} \epsilon_i a_i \right| \geq a_k - \sum_{i \in C \setminus \{k\}} a_i \geq \delta,$$

because $C \setminus \{k\} \subseteq \{1, \dots, k - 1\}$: if $k \geq 2$ then $a_k \geq \delta + \sum_{j < k} a_j \geq \delta + \sum_{i \in C \setminus \{k\}} a_i$, and if $k = 1$ then $C = \{1\}$ and $a_1 \geq \delta$.

Combining, $\text{cost}(\mu) \geq \text{cost}(C) \geq \left| \sum_{i \in C} \epsilon_i a_i \right| \geq \delta$. Since μ was arbitrary, Lemma 2.2 gives $D(M) \geq \delta$. ■

Corollary 3.2 (lower bound). *With the geometric split, whatever XY does, $D(M) \geq T_n$, so LB guarantees a share of at least*

$$\frac{1 + T_n}{2} = \frac{1}{2} \cdot \frac{2^{n+1}}{2^{n+1} - 1} = \frac{2^n}{2^{n+1} - 1}.$$

Proof. The final multiset M is a refinement of the geometric family (which is super-increasing with margin T_n) by at most $n = m - 1$ cuts (with $m = n + 1$); Theorem 3.1 applies. ■

4. Part B: Xiang Yu's counterstrategy (upper bound)

Now fix **any** composition of 1 into $m \leq n + 1$ positive pieces $a_1 \leq \dots \leq a_m$ chosen by LB. We exhibit an XY strategy ($\leq n$ cuts) forcing $D \leq T_n$.

Case $m \leq n$. XY bisects every piece: $m \leq n$ cuts. Each piece x becomes two pieces $x/2, x/2$, so every value occurs with even multiplicity in the final multiset; by Corollary 2.3, $D = 0$.

Case $m = n + 1$. This rests on two lemmas about *chains*. For a nonempty ordered tuple $(e_1, \dots, e_r)$ of positive reals define its *chain value* (nested absolute difference) by

$$v(e_1, \dots, e_r) = |\dots ||e_1 - e_2| - e_3| - \dots - e_r|.$$

A *chain* on a nonempty multiset B is a choice of an ordering $(e_1, \dots, e_r)$ of its elements, and the chain value of B for that ordering is $v(e_1, \dots, e_r)$.

Lemma 4.1 (realizability lemma). *Let P, N be disjoint finite multisets of positive reals, with sums $\Sigma P, \Sigma N$, and $P \cup N \neq \emptyset$. - (a) If $\Sigma P = \Sigma N$, then some ordering of $P \cup N$ has nested absolute difference 0. - (b) If $v := \Sigma P - \Sigma N > 0$ and no subset of $P \cup N$ has sum strictly between ΣN and ΣP , then some ordering of $P \cup N$ has nested absolute difference v .*

Proof. We construct an ordering $e_1, \dots, e_r$ of $P \cup N$ by a greedy process. Set $E_0 = 0$. At step k , while elements remain: if $E_{k-1} > 0$, append any remaining element $e_k \in N$ and put $E_k = E_{k-1} - e_k$; if $E_{k-1} < 0$, append any remaining element $e_k \in P$ and put $E_k = E_{k-1} + e_k$; if $E_{k-1} = 0$, append any remaining element from either (nonempty) side, with the corresponding sign.

We claim this process never gets stuck, i.e. the side it must draw from is never empty while the other side is nonempty. Indeed, suppose the process halts with a remaining part $P' \subseteq P$ and $N' \subseteq N$, not both empty, and let $I \subseteq P$ and $J \subseteq N$ be the parts already processed, so the current signed sum is $E = \Sigma I - \Sigma J$.

- *Halt with $E > 0$:* this requires N exhausted ($J = N$) and $P' \neq \emptyset$. Then $E = \Sigma I - \Sigma N > 0$, so $\Sigma I > \Sigma N$, while $\Sigma I = \Sigma P - \Sigma P' < \Sigma P$ (as $P' \neq \emptyset$ and all elements are positive). Hence ΣI lies strictly between ΣN and ΣP : impossible — in case (b) by the hypothesis, and in case (a) because $\Sigma I \leq \Sigma P = \Sigma N$ would give $E \leq 0$.
- *Halt with $E < 0$:* this requires P exhausted ($I = P$) and $N' \neq \emptyset$. Then the full signed sum is $\Sigma P - \Sigma N = E - \Sigma N' < 0$, contradicting $\Sigma P - \Sigma N = v > 0$ in case (b) and $\Sigma P - \Sigma N = 0$ in case (a).
- *Halt with $E = 0$:* then both sides are empty (at $E = 0$ either nonempty side may be drawn from), so nothing remains — the process is finished, not stuck.

Hence the process consumes all of $P \cup N$ and ends at $E_r = \Sigma P - \Sigma N$.

Finally we check that the nested absolute difference of the ordering $e_1, \dots, e_r$ equals $|E_r|$. Define $R_1 = e_1$ and $R_k = |R_{k-1} - e_k|$; we claim $R_k = |E_k|$ for all k . This holds for $k = 1$ since $E_1 = \pm e_1$. Inductively: if $E_{k-1} \geq 0$ and $e_k \in N$, then $E_k = E_{k-1} - e_k$ and $|E_k| = ||E_{k-1}| - e_k| = |R_{k-1} - e_k| = R_k$; if $E_{k-1} < 0$ and $e_k \in P$, then $E_k = -|E_{k-1}| + e_k$ and again $|E_k| = ||E_{k-1}| - e_k| = R_k$; if $E_{k-1} = 0$ then $|E_k| = e_k = |0 - e_k| = R_k$. Thus the chain value is $R_r = |E_r| = |\Sigma P - \Sigma N|$, which is 0 in case (a) and v in case (b). ■

Theorem 4.2 (chain theorem). *Let $a_1, \dots, a_m$ be positive reals with sum S . Then there exist a nonempty sub-multiset $B \subseteq \{a_1, \dots, a_m\}$ and an ordering $(e_1, \dots, e_r)$ of the elements of B whose chain value satisfies $v(e_1, \dots, e_r) \leq \frac{S}{2^m - 1}$.*

Proof. Consider the 2^m subset sums $s(I) = \sum_{i \in I} a_i$ ($I \subseteq \{1, \dots, m\}$); they lie in $[0, S]$. Sort them as $0 = t_0 \leq t_1 \leq \dots \leq t_{2^m-1} = S$ (with multiplicity). The $2^m - 1$ consecutive gaps sum to $t_{2^m-1} - t_0 \leq S$, so some consecutive gap satisfies

$$g := t_{j+1} - t_j \leq \frac{S}{2^m - 1}.$$

By definition of the sorted list, **no subset sum lies strictly between t_j and t_{j+1}** . Pick **distinct** subsets $A \neq B$ with $s(A) = t_{j+1}$, $s(B) = t_j$ (the sorted list has one entry per subset, so its two entries t_j, t_{j+1} come from two distinct subsets), and set $P = \{a_i : i \in A \setminus B\}$, $N = \{a_i : i \in B \setminus A\}$ (as multisets). Then $P \cup N \neq \emptyset$ and $\Sigma P - \Sigma N = s(A) - s(B) = g \geq 0$.

If $g = 0$, Lemma 4.1(a) gives an ordering of $P \cup N$ with chain value $0 \leq \frac{S}{2^m-1}$.

If $g > 0$: any subset J of $P \cup N$ with $\Sigma N < \Sigma J < \Sigma P$ would lift to the subset $J \cup (A \cap B)$ of $\{a_1, \dots, a_m\}$ whose sum lies strictly between $s(B)$ and $s(A)$, a contradiction. So the hypothesis of Lemma 4.1(b) holds, and some ordering of $P \cup N$ has chain value $g \leq \frac{S}{2^m-1}$. ■

(The bound is sharp: for $a_i = 2^{i-1}$ the subset sums are exactly $0, 1, \dots, 2^m - 1$, so every gap equals $1 = \frac{S}{2^m-1}$; and for every nonempty subset B and every ordering $(e_1, \dots, e_r)$ of B the chain value is at least 1. Indeed, an easy induction on r shows that $v(e_1, \dots, e_r) = |\sum_{i=1}^r \delta_i e_i|$ for some signs $\delta_i \in \{\pm 1\}$: if $v(e_1, \dots, e_{r-1}) = |Y|$ with Y a signed sum, then $||Y| - e_r|$ equals $|Y - e_r|$ when $Y \geq 0$ and $|-Y - e_r|$ when $Y < 0$, and both are absolute values of signed sums (negating all signs of a signed sum gives a signed sum). A signed sum of distinct powers of 2 is a nonzero integer, since the largest power present exceeds the sum of all strictly smaller powers, so the chain value is a positive integer, hence $\geq 1 = \frac{S}{2^m-1}$.)

XY's strategy from a chain. Let B be a nonempty subset of LB's pieces and $(e_1, \dots, e_r)$ an ordering of B with chain value $v := v(e_1, \dots, e_r) \leq T_n$ (Theorem 4.2 with $S = 1$, $m = n + 1$). XY plays as follows. He maintains a *running piece* of size v_{run} (initially the piece e_1 , so $v_{\text{run}} = e_1$) and processes $e_2, \dots, e_r$ in this order. To process e_k with current running value $v = v_{\text{run}}$: - if $v = 0$ (no running piece exists): declare the as-yet-untouched piece e_k the new running piece ($v_{\text{run}} := e_k$); no cut; - if $v = e_k > 0$: set the running piece and the piece e_k aside as an **equal pair** (they will cancel in D by Corollary 2.3); $v_{\text{run}} := 0$; no cut; - if $v \neq e_k$ and both are positive: cut the **larger** of the two pieces $\{v_{\text{run}}, e_k\}$ at distance $\min(v, e_k)$ from one end, producing pieces of sizes $\min(v, e_k)$ and $|v - e_k|$. Set the piece of size $\min(v, e_k)$ aside together with the smaller of the two original pieces: they form an **equal pair**. The piece of size $|v - e_k|$ becomes the new running piece; $v_{\text{run}} := |v - e_k|$. One cut.

In each case the new running value equals $|v - e_k|$, so after processing e_k the running value equals $v(e_1, \dots, e_k)$, and at the end $v_{\text{run}} = v \leq T_n$ (with no running piece left if $v = 0$).

After processing all of B , XY bisects every piece outside B ($n + 1 - r$ cuts). The total number of cuts is at most $(r - 1) + (n + 1 - r) = n$. All cut points are distinct: each cut is made inside a piece never cut before — at every moment the running piece has never been cut (it is either an as-yet-untouched original piece, namely e_1 or some e_j declared running in the $v = 0$ case, or the fresh fragment produced by the immediately preceding cut), pieces set aside are never touched again, and each e_k is processed once. Every cut lies in the interior of one of LB's original pieces (in the third case $0 < \min(v, e_k) < \max(v, e_k)$ since $v \neq e_k$ and both are positive, so the cut is interior to the piece being cut; bisections are interior too), so all of XY's marks differ from LB's marks and XY can place them all simultaneously.

In the final multiset, the pieces other than the (possible) last running piece split into disjoint pairs of equal pieces — the two halves of each bisected outside piece, each tie pair, and each equal pair formed above. Removing these pairs one at a time (Corollary 2.3) leaves the singleton $\{v\}$ if $v > 0$, and leaves $\emptyset$ if $v = 0$; in both cases

$$D(M) = v \leq T_n, \quad \text{so LB's share is at most } \frac{1 + T_n}{2} = \frac{2^n}{2^{n+1} - 1}.$$

5. Conclusion

Combining Corollary 3.2 (LB guarantees $\frac{1+T_n}{2}$ via the geometric split) with Section 4 (XY always holds LB to at most $\frac{1+T_n}{2}$), the largest value LB can guarantee is

$$c = \frac{1 + T_n}{2} = \frac{2^n}{2^{n+1} - 1}.$$

Sanity checks: $n = 1$: $c = \frac{2}{3}$ (LB marks at $\frac{1}{3}$; any cut of the long piece leaves $D \geq \frac{1}{3}$); $n = 2$: $c = \frac{4}{7}$; $n = 3$: $c = \frac{8}{15}$ — all matching exhaustive grid computation.

IMO 2026 — Problem 4

Solution by Kimi K3

2026-07-17

Source: `results/kimi-k3/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Approaches tried

- **Algebraic model of the cut.** A cut is a cevian: from a triangle with angles (A, B, C) , cutting at the vertex with angle A with parameter $\alpha \in (0, A)$ (fully at Mulan's disposal) produces halves $\{B, \alpha, 180^\circ - B - \alpha\}$ and $\{C, A - \alpha, B + \alpha\}$ (see `lemmas/cut-geometry.md`). This reduces the game to pure angle arithmetic.
- **Retrograde analysis on a discretized game** (`code/retro.py`, `approaches/retrograde-analysis.md`): computed Mulan's winning region exactly for integer-degree states. At unit 1° she wins all states iff $\theta \in \{1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90\}^\circ$ — exactly the divisors of 180° ; at unit 0.5° exactly $\{\theta : 180^\circ/\theta \in \mathbb{Z}\}$ again. This gave the conjecture.
- **Halving lemma** (`lemmas/halving-lemma.md`): for every θ , any triangle containing an angle $k\theta$ ($k \in \mathbb{Z}_{>0}$) is winning for Mulan (cut θ off $k\theta$, induct on k). So the whole question is whether Mulan can force a multiple of θ to survive a cut.

- **Mulan’s side** (`approaches/marking-strategy.md`): if $\theta = 180^\circ/n$, a one-move “round-up” cut makes *both* halves contain multiples of θ (here $180^\circ = n\theta$ being a multiple of θ is essential), then the halving lemma finishes. Win in $\leq n - 1$ moves.
- **Shan-Yu’s side** (`approaches/invariant-avoidance.md`): if θ is not of the form $180^\circ/m$, Shan-Yu maintains the invariant “no angle is a multiple of θ ” forever, starting from the equilateral triangle; the case analysis shows the invariant can only break when $180^\circ - k\theta$ is a multiple of θ , i.e. exactly when $\theta = 180^\circ/m$.
- Both strategies verified by exact rational-arithmetic simulation (`code/verify.py`): Mulan’s strategy wins exhaustively with worst case exactly $n - 1$ moves for all tested $n \leq 60$; Shan-Yu’s invariant survived 200×300 random cuts for several $\theta \neq 180^\circ/m$.
- Dead ends: naive invariants for Shan-Yu such as “all angles $< \theta$ ” (works only for $\theta > 90^\circ$) and “no angle equal to θ ” alone (not maintainable: Mulan can put θ in one half and nothing forbidden in the other); replaced by the multiple-of- θ invariant.

Current best

Answer: Mulan can guarantee victory in finitely many steps if and only if $\theta = \frac{180^\circ}{n}$ for some integer $n \geq 2$ (equivalently, iff $180^\circ/\theta$ is an integer). In that case she wins within $n - 1$ moves: with the triangle’s angles measured relative to θ , she cuts so that one angle is rounded up to the next multiple of θ ; because $180^\circ = n\theta$ the supplementary angle at the cut is then also a multiple of θ , so *both* halves contain a multiple of θ ; then she repeatedly cuts θ off the surviving multiple ($k\theta \rightarrow (k-1)\theta$) until an angle exactly θ appears in every branch. For all other θ , Shan-Yu starts with an equilateral triangle and always keeps a half having no angle equal to an integer multiple of θ (one exists among the two halves of any cut, precisely because $\theta \neq 180^\circ/m$), so Mulan can never win.

Full proof

Setup and angle bookkeeping

Throughout, describe a triangle by its multiset of angles (A, B, C) , positive reals with $A + B + C = 180^\circ$. The game only depends on the angles.

Lemma 0 (the cut). *Suppose the current triangle has angles A at vertex X , B at Y , C at Z . Mulan’s point P lies on some side; say it lies on YZ (the opposite vertex is then X). Put $\alpha = \angle YXP$. Then the two halves have angle multisets*

$$H_1 = \{B, \alpha, 180^\circ - B - \alpha\}, \quad H_2 = \{C, A - \alpha, B + \alpha\},$$

and every value $\alpha \in (0, A)$ is achieved by a suitable interior point P of YZ .

Proof. See `lemmas/cut-geometry.md`. (Angle chase with $A + B + C = 180^\circ$; the range of α by continuity of $P \mapsto \angle YXP$ and the intermediate value theorem.) ■

So a move of Mulan consists of: choosing which angle of (A, B, C) is the *cut angle* A , choosing which of the remaining angles is B (the one that ends up in the same half as α), and choosing a real number $\alpha \in (0, A)$. Then Shan-Yu replaces the triangle by one of H_1, H_2 .

Call an angle **marked** if it is an integer multiple of θ , i.e. equal to $k\theta$ for some integer $k \geq 1$ (necessarily $k\theta < 180^\circ$ since it is an angle of a triangle).

The halving lemma

Lemma 1. *Fix any θ . If the current triangle has a marked angle $k\theta$ ($k \geq 1$ integer), then Mulan can force victory from it within $k - 1$ moves.*

Proof. Induction on k . If $k = 1$ the triangle has an angle θ and the game is already over. If $k \geq 2$, let the angles be $A = k\theta$, B , C ; Mulan cuts at A with $\alpha = \theta \in (0, A)$. By Lemma 0 the halves are

$$H_1 = \{B, \theta, 180^\circ - B - \theta\}, \quad H_2 = \{C, (k-1)\theta, B + \theta\},$$

both genuine triangles since $B < 180^\circ - 2\theta$. If Shan-Yu keeps H_1 , the game stops next (angle θ present); if he keeps H_2 , Mulan wins from it within $k-2$ further moves by induction. Total $\leq k-1$ moves. ■

Theorem 1: $\theta = 180^\circ/n$ — Mulan wins

Theorem 1. *Let $\theta = 180^\circ/n$ for an integer $n \geq 2$. Then Mulan has a strategy that wins within $n - 1$ moves from every initial triangle.*

Proof. If the initial triangle has a marked angle $k\theta$ ($1 \leq k \leq n-1$), Lemma 1 wins within $k-1 \leq n-2$ moves. So assume no angle of $\mathcal{T} = (A, B, C)$ is marked.

For an unmarked angle x define

$$d(x) = (\lfloor x/\theta \rfloor + 1)\theta - x \in (0, \theta),$$

the distance from x up to the nearest multiple of θ above it.

Claim 1. $d(A) + d(B) + d(C) \in \{\theta, 2\theta\}$.

Proof. Each $x + d(x)$ is a multiple of θ , and $(A + d(A)) + (B + d(B)) + (C + d(C)) = 180^\circ + d(A) + d(B) + d(C) = n\theta + d(A) + d(B) + d(C)$, so the sum of the d 's is a multiple of θ ; as each lies in $(0, \theta)$, the sum lies in $(0, 3\theta)$, hence equals θ or 2θ . ■

Claim 2. *There exist two distinct angles u, v of $\mathcal{T}$ with $d(u) < v$.*

Proof. Suppose not. Then in particular $d(A) \geq B$, $d(B) \geq C$, $d(C) \geq A$, and summing gives

$$d(A) + d(B) + d(C) \geq A + B + C = 180^\circ = n\theta.$$

By Claim 1 the left side is at most 2θ , so $n\theta \leq 2\theta$, i.e. $n = 2$, and all three inequalities must be equalities: $d(A) = B$, $d(B) = C$, $d(C) = A$. But then $A + d(A) = A + B$ is a multiple of θ by definition of $d(A)$, while $A + B = 180^\circ - C = 2\theta - C$ (here $n = 2$). Since $0 < A + B < 180^\circ$ we get $A + B = \theta$, hence $C = \theta$, contradicting that C is unmarked. ■

Mulan's move. Pick a pair $u \neq v$ with $d(u) < v$ (Claim 2) and let w be the third angle. Mulan cuts at the vertex with angle v , putting u in the role of B and

$$\alpha = d(u) \in (0, v).$$

Let $u + d(u) = k\theta$ ($k \geq 1$ integer, a multiple of θ just above u). Since $k\theta = u + \alpha < u + v < 180^\circ = n\theta$, we have $1 \leq k \leq n-1$. By Lemma 0 the halves are

$$H_1 = \{u, d(u), 180^\circ - u - d(u)\}, \quad H_2 = \{w, v - d(u), u + d(u)\}.$$

In H_2 the angle $u + d(u) = k\theta$ is marked. In H_1 the angle

$$180^\circ - u - d(u) = n\theta - k\theta = (n - k)\theta$$

is marked as well, because $180^\circ = n\theta$ is a multiple of θ and $1 \leq n - k \leq n - 1$. (This is the only place where the hypothesis $\theta = 180^\circ/n$ is used.)

So **both** halves contain a marked angle. Whichever half Shan-Yu keeps, the new triangle contains some marked angle $j\theta$ with $1 \leq j \leq n - 1$, and by Lemma 1 Mulan wins from it within $j - 1 \leq n - 2$ further moves.

Total: at most $1 + (n - 2) = n - 1$ moves. ■

Theorem 2: $\theta \neq 180^\circ/m$ — **Shan-Yu avoids forever**

Theorem 2. Suppose $\theta \neq 180^\circ/m$ for every integer $m \geq 2$. Then Shan-Yu has a strategy (choice of initial triangle and of which half to discard) ensuring that no triangle in the game ever has an angle θ ; in particular Mulan cannot guarantee a win.

Proof. Shan-Yu maintains the invariant:

(I) no angle of the current triangle is marked (i.e. no angle is an integer multiple of θ).

Since θ itself is a multiple of θ , invariant (I) implies the game never stops, so Mulan never wins.

Initial triangle. Shan-Yu builds the equilateral triangle $(60^\circ, 60^\circ, 60^\circ)$. If $60^\circ = k\theta$ for an integer $k \geq 1$, then $\theta = 60^\circ/k = 180^\circ/(3k)$, which is of the excluded form $180^\circ/m$ ($m = 3k \geq 3$). Hence (I) holds initially.

Maintenance. Assume $\mathcal{T} = (A, B, C)$ satisfies (I) and Mulan cuts at A with some $\alpha \in (0, A)$ and some choice of B , producing $H_1 = \{B, \alpha, 180^\circ - B - \alpha\}$ and $H_2 = \{C, A - \alpha, B + \alpha\}$ (Lemma 0). We show at least one half satisfies (I); Shan-Yu keeps it.

Case 1: $\alpha = k\theta$ is a multiple of θ . Then H_2 satisfies (I): C is unmarked by hypothesis; if $A - \alpha = j\theta$ for an integer $j \geq 1$ (note $A - \alpha > 0$), then $A = (j + k)\theta$ is marked, contradiction; if $B + \alpha = j\theta$ for an integer $j \geq 1$, then $B = (j - k)\theta$ with $j - k \geq 1$ (since $B > 0$), so B is marked, contradiction.

Case 2: α is not a multiple of θ , and $180^\circ - B - \alpha$ is not a multiple of θ . Then H_1 satisfies (I): B is unmarked by hypothesis, and the other two angles of H_1 are non-multiples by the case assumption.

Case 3: α is not a multiple of θ , but $180^\circ - B - \alpha = k\theta$ for some integer $k \geq 1$. (Note $0 < k\theta < 180^\circ$, since it is an angle of H_1 .) Then H_2 satisfies (I):

- C is unmarked by hypothesis.
- $A - \alpha = A - (180^\circ - B - k\theta) = k\theta - C$. If $A - \alpha = j\theta$ for an integer $j \geq 1$ (it is positive), then $C = (k - j)\theta$ with $k - j \geq 1$ (since $C > 0$), so C is marked — contradiction.
- $B + \alpha = 180^\circ - k\theta$. If $B + \alpha = j\theta$ for an integer $j \geq 1$ (it is positive), then $180^\circ = (j + k)\theta$ with $j + k \geq 2$, i.e. $\theta = 180^\circ/(j + k)$ is of the excluded form — contradiction.

The three cases are exhaustive, so the invariant can always be maintained, and the proof is complete. ■

Conclusion

By Theorem 1, if $\theta = 180^\circ/n$ for an integer $n \geq 2$, Mulan guarantees victory within $n - 1$ moves against any play of Shan-Yu (any initial triangle, any discards). By Theorem 2, for every other $\theta \in (0^\circ, 180^\circ)$ Shan-Yu prevents Mulan from ever winning. Hence

$\text{Mulan can guarantee victory} \iff \theta = \frac{180^\circ}{n} \text{ for some integer } n \geq 2 \iff \frac{180^\circ}{\theta} \in \mathbb{Z}.$

Explicitly: $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, \frac{180^\circ}{7}, 22.5^\circ, 20^\circ, \dots\}$.

Verification. The discretized game (1° and 0.5° grids, `code/retro.py`) yields exactly this family; the explicit strategies above were verified with exact rational arithmetic (`code/verify.py`): for $\theta = 180^\circ/n$ Mulan's strategy wins in exactly at most $n - 1$ moves (worst case = $n - 1$, tight), exhaustively over Shan-Yu's choices, for all tested $n \leq 90$ on random rational triangles; the pairing Claim 2 and the round-up move were additionally checked exhaustively on $\sim 6 \times 10^4$ unmarked grid triples for each $n \mid 360$, $n \leq 30$ (`code/check_pairing.py`); for $\theta \neq 180^\circ/m$ Shan-Yu's invariant survived 200×300 random cuts for each tested θ .

IMO 2026 — Problem 5

Solution by Kimi K3

2026-07-17

Source: `results/kimi-k3/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Approaches tried

- Tried the first guesses $f \equiv c$ and power functions $f(x) = x^a$. Constants fail immediately by letting $y \rightarrow 0^+$ in the right-hand inequality; non-identity powers fail by homogeneity. This was a dead end for the full classification, but it suggested looking for functions making $f(x) + y$ equal to $x + f(y)$.
- Checked $f(x) = x + c$. It works because then $f(x) + y = x + f(y)$, so the middle term is exactly the arithmetic mean of the two numbers x and $f(y)$ occurring in the quadratic and geometric means. This gave the candidate family and shifted the problem to proving that $f(x) - x$ is constant.
- Developed the successful “mean interval” approach (see `approaches/mean-interval.md`): for fixed x, y , the two given inequalities say that $(f(x) + y)^2$ lies in the interval $[(x + f(y))^2 - (x - f(y))^2, (x + f(y))^2 + (x - f(y))^2]$. Substituting $x = f(y)$ forces $f(f(y)) = 2f(y) - y$.
- Developed the successful “orbit/forbidden interval” argument (see `approaches/orbit-forbidden-interval.md`): the relation $f(f(t)) = 2f(t) - t$ makes forward iterates arithmetic progressions and positivity forces $f(t) \geq t$. If the increment $f(t) - t$ took two values $c > d$, then the c -orbit eventually falls inside an interval where the left-hand mean inequality with a d -increment point is impossible.

Current best

The complete set of solutions is

$$\boxed{f(x) = x + c \quad (x > 0)}$$

for an arbitrary constant $c \geq 0$. The proof is complete below: setting $M = f(x) + y$, $N = x + f(y)$ rewrites the hypothesis as $|M^2 - N^2| \leq (x - f(y))^2$; taking $x = f(y)$ gives $f(f(y)) = 2f(y) - y$, whose iterates are arithmetic and positivity gives $p(x) := f(x) - x \geq 0$. A descent argument shows that if $p(x) = c > 0$, then the orbit of x under f starts at some $r \in (0, c]$ and is $r + nc$. If p took two values $c > d$, the left-hand inequality would fail on a forbidden interval whose length can be made larger than the spacing c of the c -orbit, forcing a point of that orbit into the forbidden interval. Hence p is constant.

Full proof

We will prove that the solutions are exactly

$$f(x) = x + c \quad (x > 0)$$

for constants $c \geq 0$.

For fixed $x, y > 0$, put

$$v = f(y), \quad M = f(x) + y, \quad N = x + v = x + f(y).$$

Since all quantities are positive, the right-hand given inequality is equivalent to

$$M^2 \geq 4xv.$$

But

$$4xv = (x + v)^2 - (x - v)^2 = N^2 - (x - f(y))^2.$$

Similarly, the left-hand given inequality is equivalent to

$$M^2 \leq 2(x^2 + v^2) = (x + v)^2 + (x - v)^2 = N^2 + (x - f(y))^2.$$

Thus the whole hypothesis is equivalent to

$$\boxed{|(f(x) + y)^2 - (x + f(y))^2| \leq (x - f(y))^2} \quad (1)$$

for all $x, y > 0$. This is the key mean estimate recorded in `lemmas/key-mean-estimate.md`.

Now substitute $x = f(y)$ in (1). The right side is 0, so

$$(f(f(y)) + y)^2 = (2f(y))^2.$$

Both sides are positive, hence

$$\boxed{f(f(y)) = 2f(y) - y} \quad (2)$$

for every $y > 0$.

Let $f^{[0]}(t) = t$ and $f^{[n+1]}(t) = f(f^{[n]}(t))$. Fix t and write

$$p(t) = f(t) - t.$$

Applying (2) repeatedly gives, with $a_n = f^{[n]}(t)$,

$$a_{n+2} = f(f(a_n)) = 2f(a_n) - a_n = 2a_{n+1} - a_n.$$

Since $a_0 = t$ and $a_1 = t + p(t)$, induction yields

$$\boxed{f^{[n]}(t) = t + np(t)} \quad (n \geq 0). \quad (3)$$

Every $f^{[n]}(t)$ is positive. If $p(t) < 0$, then $t + np(t) < 0$ for all sufficiently large n , contradicting (3). Therefore

$$\boxed{p(t) = f(t) - t \geq 0} \quad (t > 0). \quad (4)$$

Also (3) shows that p is constant on each forward orbit of f ; this is `lemmas/iterate-positivity.md`.

We next need a descent statement. Suppose $p(t) = c$ and $t > c > 0$. Put $z = t - c > 0$ and write $e = p(z) \geq 0$. Then

$$f(t) = t + c, \quad f(z) = z + e = t - c + e.$$

Apply (1) to the pair (t, z) . We have

$$f(t) + z = (t + c) + (t - c) = 2t,$$

and

$$t + f(z) = t + (t - c + e) = 2t - c + e,$$

while

$$t - f(z) = t - (t - c + e) = c - e.$$

Hence (1) gives

$$|c - e| (4t - c + e) \leq (c - e)^2. \quad (5)$$

If $c \neq e$, dividing (5) by $|c - e|$ gives

$$4t - c + e \leq |c - e|.$$

But $t > c$, so

$$4t - c + e > 3c + e > |c - e|,$$

where the last inequality is immediate both when $e \geq c$ and when $e < c$. This contradiction proves $e = c$. Consequently

$$f(z) = z + c = t.$$

So, whenever $p(t) = c$ and $t > c$, the point $t - c$ is still positive, has the same increment c , and maps to t ; this is `lemmas/descent.md`.

It follows that for any t with $p(t) = c > 0$, repeatedly subtracting c while the current point remains $> c$ is legitimate and preserves the increment c . After finitely many subtractions we reach a number

$$r \in (0, c]$$

with $p(r) = c$. Therefore, by (3),

$$\boxed{f^{[n]}(r) = r + nc \quad (n \geq 0), \quad p(r + nc) = c.} \quad (6)$$

We now prove that p is constant on all of $\mathbb{R}_{>0}$; this is the constancy argument in `lemmas/constancy.md`. Suppose not. Then there exist points whose increments are

$$c > d \geq 0.$$

By the previous paragraph, choose $r \in (0, c]$ with $p(r) = c$, and define

$$x_n = r + nc \quad (n \geq 0),$$

so that $f(x_n) = x_n + c$ by (6).

Choose a point y with $p(y) = d$. If $d > 0$, replace y by a suitable forward iterate $f^{[m]}(y) = y + md$; by (3) this keeps $p(y) = d$ and makes y arbitrarily large. Hence in all cases we may assume that

$$\boxed{2s > c, \quad \text{where } s := \sqrt{2(c-d)(c+d+2y)}}. \quad (7)$$

Indeed, if $d > 0$ this is achieved by taking the above iterate large enough; if $d = 0$, then already

$$s^2 = 2c(c+2y) > 2c^2,$$

so $s > \sqrt{2}c$ and certainly $2s > c$.

For this fixed y , we have $f(y) = y + d$. Apply the left-hand part of the original inequality with $x = x_n$:

$$\sqrt{\frac{x_n^2 + (y+d)^2}{2}} \geq \frac{f(x_n) + y}{2} = \frac{x_n + c + y}{2}.$$

Squaring gives

$$2(x_n^2 + (y+d)^2) \geq (x_n + c + y)^2. \quad (8)$$

Consider the quadratic

$$Q(X) = 2(X^2 + (y+d)^2) - (X + c + y)^2.$$

With $A = c + y$, a direct expansion gives

$$Q(X) = (X - A)^2 - s^2,$$

because

$$s^2 = 2((c+y)^2 - (y+d)^2) = 2(c-d)(c+d+2y).$$

Thus (8) says exactly that x_n must not lie in the open interval

$$I = (A - s, A + s).$$

But $A + s = c + y + s > c \geq r = x_0$, so $x_0 < A + s$. If $x_0 > A - s$, then already $x_0 \in I$, contradicting (8) for $n = 0$. Otherwise $x_0 \leq A - s$. Since $x_n \rightarrow \infty$ and the step of the sequence (x_n) is $c < 2s$ by (7), the sequence cannot jump over the interval I : if n is the first index with $x_n > A - s$, then $n \geq 1$, $x_{n-1} \leq A - s$, and

$$x_n = x_{n-1} + c \leq A - s + c < A - s + 2s = A + s.$$

Hence $x_n \in I$, again contradicting (8). Therefore the assumed two increments $c > d$ cannot occur.

So $p(t)$ is a constant $c \geq 0$, i.e.

$$f(t) = t + c \quad (t > 0).$$

It remains to verify that every such function works. Let $c \geq 0$ and $f(t) = t + c$. For $x, y > 0$, set

$$a = x, \quad b = f(y) = y + c.$$

Then

$$f(x) + y = x + c + y = a + b.$$

Consequently the required inequalities become

$$\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2} \geq \sqrt{ab}.$$

The right inequality is $(a + b)^2 \geq 4ab$, and the left inequality is $2(a^2 + b^2) \geq (a + b)^2$; both follow from $(a - b)^2 \geq 0$. Hence all functions $f(x) = x + c$ with $c \geq 0$ satisfy the condition, and no other functions do.

Therefore the answer is

$f(x) = x + c \ (x > 0) \text{ for some constant } c \geq 0.$

IMO 2026 — Problem 6

Solution by Kimi K3

2026-07-19

Source: `results/kimi-k3/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved — complete proof written below. The previous gap (Step 4 applied self-duality to a provably-infinite minimal transversal while self-duality was only proved for finite ones) is now closed by a new **Compactness Theorem** (Section “Step 4a”), proved from the greedy *killing property* (K) via a descent + pigeonhole argument. No infinite minimal transversals can exist, so the finiteness of the permanent family $\mathcal{C}$ follows from the peeling argument using only *finite* self-duality. All steps verified line by line; key new number-theoretic fact (K) double-checked numerically in `code/check_Kprime.py`.

Approaches tried

- **Minimal-hitting-set (transversal) framework** (session 1): $S_n = \{m : \gcd(m, a_i) > 1 \ \forall i \leq n\}$; permanents = minimal elements of S_∞ ; reduction of the problem to finiteness of the set of permanents. Empirics: permanent family always finite; $a_{n+T} = a_n + L$ verified for 100+ values of a_1 .
- **Self-dual intersecting clutter** (session 2): permanents, viewed as sets of primes, form an antichain $\mathcal{C}$ that is pairwise intersecting and self-dual *for finite transversals*. Abstract finiteness theorem by peeling — but its last step needed self-duality for infinite minimal transversals: **GAP** (confirmed by independent review).
- **Compactness via the killing property (THIS SESSION — successful)**. The greedy rule gives a *killing property* (K): any finite prime-set F with $\pi(F) > a_1$ containing no

permanent is avoided by some permanent of smaller product. Using (K) we prove $\mathcal{C}$ has **no infinite minimal transversal** (descent on π through private members, then pigeonhole on $\pi(t^* \setminus \{w\}) \leq a_1$, then finite self-duality gives a member strictly inside another member — contradicting the antichain property). With compactness, the peeling’s “runs forever” case is impossible, and the “stops” case contradicts antichain + finite self-duality. Hence $\mathcal{C}$ is finite and Step 2 concludes.

Current best

Full proof below (final). Final statement proved: with $\mathcal{M}$ = minimal elements of S_∞ (finite by the proof), $L = \text{lcm}(\mathcal{M})$ and $T = \#(S_\infty \cap [a_1, a_1 + L))$, one has $a_{n+T} = a_n + L$ for all $n \geq 1$.

Full proof

Notation. For an integer $m \geq 2$, $\text{supp}(m)$ = set of prime divisors of m . For a finite set s of primes, $\pi(s) := \prod_{p \in s} p$ (a squarefree integer; $\pi(\emptyset) = 1$). A set s is a *transversal* of a family $\mathcal{F}$ if $s \cap F \neq \emptyset$ for every $F \in \mathcal{F}$; a transversal is *minimal* if no proper subset is a transversal.

Step 1: The sequence enumerates $S_\infty \cap [a_1, \infty)$

Let $S_\infty := \{m \geq 1 : \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}$.

Claim. $(a_n)_{n \geq 1}$ is the increasing enumeration of $S_\infty \cap [a_1, \infty)$.

Proof. Every term lies in S_∞ : for $i < j$, the defining property of a_j (smallest integer $> a_{j-1}$ with $\gcd(a_j, a_{i'}) > 1$ for all $i' \leq j-1$) applied with $i' = i$ gives $\gcd(a_j, a_i) > 1$; also $\gcd(a_i, a_i) = a_i > 1$. So any two terms share a factor > 1 , i.e. every term is in S_∞ ; also $a_n \geq a_1$ as the sequence is strictly increasing.

Conversely let $x \in S_\infty$, $x \geq a_1$, and suppose x is not a term. Then $x > a_1$, and since the terms form an infinite strictly increasing sequence of integers, there is a unique n with $a_n < x < a_{n+1}$. By definition a_{n+1} is the smallest integer $> a_n$ in $S_n := \{m : \gcd(m, a_i) > 1 \forall i \leq n\}$. But $x \in S_\infty \subseteq S_n$ and $x > a_n$, so $a_{n+1} \leq x$ — contradiction. ■

Step 2: Reduction to finiteness of the minimal elements of S_∞

S_∞ is closed under taking multiples. Let $\mathcal{M}$ be the set of its minimal elements under divisibility. Every $m \in S_\infty$ is a multiple of some element of $\mathcal{M}$: among the divisors of m lying in S_∞ (a nonempty finite set) the smallest is a minimal element. Hence $S_\infty = \bigcup_{h \in \mathcal{M}} h\mathbb{Z}^+$.

Claim. If $\mathcal{M}$ is finite, the problem’s conclusion holds.

Proof. Write $\mathcal{M} = \{h_1, \dots, h_k\}$ and $L := \text{lcm}(h_1, \dots, h_k) \geq 1$. Since each $h_i \mid L$, for every positive integer m : $m \in S_\infty \iff \text{some } h_i \mid m \iff \text{some } h_i \mid m + L \iff m + L \in S_\infty$. Let $T := \#(S_\infty \cap [a_1, a_1 + L))$; since $a_1 \in S_\infty$ (Step 1), $T \geq 1$. The map $x \mapsto x + L$ is an order-preserving bijection $S_\infty \cap [a_1, \infty) \rightarrow S_\infty \cap [a_1 + L, \infty)$; iterating, each block $[a_1 + jL, a_1 + (j+1)L)$ ($j \geq 0$) contains exactly T elements of S_∞ , namely $a_1 + jL, \dots, a_T + jL$. By Step 1 the terms in that block are exactly $a_{jT+1}, \dots, a_{jT+T}$, so $a_{jT+r} = a_r + jL$ for $1 \leq r \leq T$, $j \geq 0$. Writing $n = jT + r$: $a_{n+T} = a_{(j+1)T+r} = a_r + (j+1)L = a_n + L$ for every $n \geq 1$. ■

Step 3: The minimal elements form an intersecting, finite-self-dual antichain; the killing property

Every $h \in \mathcal{M}$ is squarefree: if $p^2 \mid h$ for a prime p , then $h/p \in S_\infty$ (since $\text{supp}(h/p) = \text{supp}(h)$ still meets every $\text{supp}(a_i)$), contradicting minimality. So each $h \in \mathcal{M}$ has the form $h = \pi(t)$ for a finite set of primes $t = \text{supp}(h)$. For squarefree $h = \pi(t)$ we have $h \in S_\infty$ iff t is a transversal of $\{\text{supp}(a_i) : i \geq 1\}$, and h minimal under divisibility iff t is minimal under inclusion. Hence

$$\mathcal{C} := \{\text{supp}(h) : h \in \mathcal{M}\}$$

is exactly the family of minimal (under inclusion) transversals of $\{\text{supp}(a_i) : i \geq 1\}$. It is an antichain of finite sets: if $t \subseteq t'$ are both minimal transversals, then $t = t'$.

Claim 3.1 (intersecting). Any two members of $\mathcal{C}$ intersect.

Proof. Let $t \in \mathcal{C}$. Every multiple of $\pi(t)$ lies in S_∞ (its support contains t , hence meets every $\text{supp}(a_i)$). Choose $k \geq 1$ with $\pi(t)^k \geq a_1$; by Step 1, $\pi(t)^k$ is a term a_j , and $\text{supp}(a_j) = t$. Every $t' \in \mathcal{C}$ is a transversal of $\{\text{supp}(a_i)\}$, so $t' \cap t = t' \cap \text{supp}(a_j) \neq \emptyset$. ■

Claim 3.2 (self-dual for finite transversals). Every FINITE minimal transversal of the family $\mathcal{C}$ belongs to $\mathcal{C}$.

Proof. Let U be a FINITE minimal (under inclusion) transversal of $\mathcal{C}$. For each i , $a_i \in S_\infty$ is a multiple of some $\pi(t)$ with $t \in \mathcal{C}$; then $t \subseteq \text{supp}(a_i)$, and U meeting t implies U meets $\text{supp}(a_i)$. Hence $\pi(U) \in S_\infty$, so $\pi(U)$ is a multiple of some $\pi(t')$ with $t' \in \mathcal{C}$, i.e. $t' \subseteq U$. By Claim 3.1 t' meets every member of $\mathcal{C}$, so t' is a transversal of $\mathcal{C}$ contained in U ; minimality of U gives $U = t' \in \mathcal{C}$. ■

Claim 3.3 (the killing property). **(K)** Every integer $x > a_1$ with $x \notin S_\infty$ is coprime to some term $a_i < x$.

Proof. By Step 1, x is not a term; since the terms increase to infinity and $x > a_1$, there is n with $a_n < x < a_{n+1}$. If $\gcd(x, a_i) > 1$ for all $i \leq n$, then $x \in S_n$ and $x > a_n$, so by the greedy minimality of a_{n+1} we'd have $a_{n+1} \leq x$ — contradiction. Hence $\gcd(x, a_i) = 1$ for some $i \leq n$, and $a_i \leq a_n < x$. ■

Claim 3.4 (K — killing property for finite sets). Let F be a finite set of primes with $\pi(F) > a_1$ such that F contains no member of $\mathcal{C}$. Then there exists $t' \in \mathcal{C}$ with $t' \cap F = \emptyset$ and $\pi(t') < \pi(F)$.

Proof. If $\pi(F) \in S_\infty$, then by Step 2's descent some $h = \pi(t) \in \mathcal{M}$ divides $\pi(F)$, so $t \subseteq F$ is a member contained in F — excluded. Hence $\pi(F) \notin S_\infty$, and $\pi(F) > a_1$. By (K) there is a term $a_i < \pi(F)$ with $\gcd(a_i, \pi(F)) = 1$. Since $a_i \in S_\infty$, some $h' = \pi(t') \in \mathcal{M}$ divides a_i ($t' \in \mathcal{C}$); then $\pi(t') \leq a_i < \pi(F)$, and $\gcd(\pi(t'), \pi(F)) \mid \gcd(a_i, \pi(F)) = 1$, i.e. $t' \cap F = \emptyset$. ■

Step 4a: The Compactness Theorem — no infinite minimal transversals

Theorem. Every transversal of $\mathcal{C}$ contains a finite transversal. Equivalently, $\mathcal{C}$ has no infinite minimal transversal.

Proof. We prove the first form; the equivalence is explained at the end. Let W be a transversal of $\mathcal{C}$; we show that W contains a finite transversal. If W is finite there is nothing to prove, so assume W infinite. Enumerate its elements as $w_1, w_2, \dots$ (possible since W is a set of primes, hence

countable). Put $W_0 := W$ and, for $n \geq 1$,

$$W_n := \begin{cases} W_{n-1} \setminus \{w_n\}, & \text{if this is still a transversal of } \mathcal{C}, \\ W_{n-1}, & \text{otherwise.} \end{cases}$$

Each W_n is a transversal by induction. Let $T := \bigcap_{n \geq 0} W_n$. For each $e \in \mathcal{C}$, the sets $e \cap W_n$ are nonempty (as W_n is a transversal), decreasing in n , and contained in the finite set e ; hence they stabilize, and $e \cap T = \bigcap_n (e \cap W_n) \neq \emptyset$. So T is a transversal contained in W . Moreover T is minimal: if $w \in T$, say $w = w_n$, then w survived stage n , i.e. $W_{n-1} \setminus \{w\}$ is not a transversal; so some $e_w \in \mathcal{C}$ is disjoint from $W_{n-1} \setminus \{w\}$, while $e_w \cap W_{n-1} \neq \emptyset$; hence $e_w \cap W_{n-1} = \{w\}$, and since $T \subseteq W_{n-1}$ and $w \in T$,

$$e_w \cap T = \{w\} \quad (\text{a private member for } w).$$

Now suppose, for contradiction, that W contains no finite transversal. Then T cannot be finite (it would itself be a finite transversal inside W), so T is an infinite transversal in which every point has a private member. We show this is impossible. (For the equivalence stated in the theorem: an infinite minimal transversal would be a transversal containing no finite transversal, since any finite sub-transversal would contradict minimality; and the construction above produces, inside any transversal without a finite sub-transversal, an infinite minimal one.)

Descent. Fix $w \in T$. We construct members $M_0, M_1, \dots, M_k$ of $\mathcal{C}$, all containing w , with strictly decreasing products, ending with $\pi(M_k) \leq a_1 w$. Set $M_0 := e_w$ (the private member, $w \in M_0$). Suppose M_j is constructed with $w \in M_j$. If $\pi(M_j) \leq a_1 w$, stop. Otherwise apply (K) to $F := M_j \setminus \{w\}$: - F contains no member of $\mathcal{C}$: a member $e \subseteq F \subsetneq M_j$ would contradict the antichain property (both $e, M_j \in \mathcal{C}$); - $\pi(F) = \pi(M_j)/w > a_1$.

So (K) yields $M_{j+1} \in \mathcal{C}$ disjoint from $M_j \setminus \{w\}$ with $\pi(M_{j+1}) < \pi(M_j)/w$. By Claim 3.1, $M_{j+1} \cap M_j \neq \emptyset$, and the intersection is disjoint from $M_j \setminus \{w\}$; hence $w \in M_{j+1}$. Since $w \geq 2$, $\pi(M_{j+1}) < \pi(M_j)/2$, so the positive integers $\pi(M_j)$ strictly decrease and the process stops at some $M_k =: t_w^*$ with

$$w \in t_w^*, \quad \pi(t_w^*) \leq a_1 w.$$

Pigeonhole. For each $w \in T$, the set $t_w^* \setminus \{w\}$ is a set of primes with $\pi(t_w^* \setminus \{w\}) = \pi(t_w^*)/w \leq a_1$; every such set is a subset of the (finite) set of primes $\leq a_1$, so there are only finitely many possibilities. Since T is infinite, some fixed finite set of primes P_0 (with $\pi(P_0) \leq a_1$) satisfies

$$t_w^* \setminus \{w\} = P_0 \quad \text{for infinitely many distinct } w \in T; \text{ pick such } w_1, w_2, w_3, \dots$$

Note $w_j \notin P_0$ and $t_{w_j}^* = P_0 \cup \{w_j\}$.

Every member meets P_0 . Let $e \in \mathcal{C}$. By Claim 3.1, e meets each $t_{w_j}^* = P_0 \cup \{w_j\}$. If $e \cap P_0 = \emptyset$, then $e \cap t_{w_j}^* = e \cap \{w_j\}$, so $w_j \in e$ for every $j = 1, 2, \dots$ — but the w_j are infinitely many distinct primes and e is finite: contradiction. Hence $e \cap P_0 \neq \emptyset$ for every $e \in \mathcal{C}$, i.e. P_0 is a transversal of $\mathcal{C}$.

Contradiction. P_0 is a finite transversal; successively deleting redundant elements yields a minimal transversal $U \subseteq P_0$, still finite. By Claim 3.2, $U \in \mathcal{C}$. Now $U \subseteq P_0$ and $w_1 \notin P_0$ give $w_1 \notin U$, while $w_1 \in t_{w_1}^*$; hence

$$U \subsetneq P_0 \cup \{w_1\} = t_{w_1}^*,$$

two members of $\mathcal{C}$, one properly containing the other — contradicting the antichain property. ■

Step 4b: Finiteness of $\mathcal{C}$ — the peeling argument

Theorem. $\mathcal{C}$ is finite.

Proof. Suppose $\mathcal{C}$ is infinite. Fix a member $t \in \mathcal{C}$. By Claim 3.1 every member meets the finite set t , so

$$\mathcal{C} = \bigsqcup_{\emptyset \neq I \subseteq t} \mathcal{C}_I, \quad \mathcal{C}_I := \{e \in \mathcal{C} : e \cap t = I\},$$

a finite union; fix I with $\mathcal{C}_I$ infinite.

Observation (b). If $g \in \mathcal{C}$ is disjoint from I , then g meets every member of $\mathcal{C}_I$ at a point outside t : indeed $g \cap e \neq \emptyset$ (Claim 3.1) and $g \cap e \cap t \subseteq g \cap (e \cap t) = g \cap I = \emptyset$.

Fix an enumeration of the countable family $\mathcal{C}$. Set $Z_0 := \emptyset$ and $\mathcal{F}_0 := \mathcal{C}_I$ (infinite). For $s = 0, 1, 2, \dots$, assuming $\mathcal{F}_s := \{e \in \mathcal{C}_I : e \supseteq Z_s\}$ is infinite, do the following (stage $s + 1$):

Case A: no member of $\mathcal{C}$ is disjoint from $I \cup Z_s$. Then $I \cup Z_s$ is a transversal of $\mathcal{C}$; it is finite, so deleting redundant elements yields a finite minimal transversal $U \subseteq I \cup Z_s$, and $U \in \mathcal{C}$ by Claim 3.2. Every $e \in \mathcal{F}_s$ satisfies $e \supseteq I \cup Z_s \supseteq U$; by the antichain property $e = U$. Hence $\mathcal{F}_s \subseteq \{U\}$, contradicting that $\mathcal{F}_s$ is infinite. **So Case A cannot occur.**

Case B: some member is disjoint from $I \cup Z_s$. Let g_s be such a member of least index in the enumeration. Since $g_s \cap I = \emptyset$, observation (b) says g_s meets every $e \in \mathcal{F}_s$ outside t . The set $g_s \setminus t$ is finite and $\mathcal{F}_s$ is infinite, so by the pigeonhole principle some point $r \in g_s \setminus t$ lies in infinitely many members of $\mathcal{F}_s$. Set $z_{s+1} := r$ (note $r \notin Z_s$ since $g_s \cap Z_s = \emptyset$, and $r \notin t$) and $\mathcal{F}_{s+1} := \{e \in \mathcal{F}_s : z_{s+1} \in e\}$, which is infinite.

Since Case A never occurs, the process runs forever and produces distinct primes $Z_\omega := \{z_1, z_2, \dots\}$, all outside t .

Claim (c). $W := I \cup Z_\omega$ is a transversal of $\mathcal{C}$ with no finite sub-transversal.

Proof of (c). Suppose some $g \in \mathcal{C}$ is disjoint from W . Then g is disjoint from $I \cup Z_s$ for every s , so g is eligible at every stage: $\text{index}(g_s) \leq \text{index}(g)$ for all s . But g_{s+1} , being disjoint from $I \cup Z_{s+1} \supseteq I \cup Z_s$, was already eligible at stage $s + 1$, so $\text{index}(g_{s+1}) \geq \text{index}(g_s)$; equality would force $g_{s+1} = g_s$, impossible since $z_{s+1} \in g_s$ while $g_{s+1} \cap Z_{s+1} = \emptyset$. Hence $(\text{index}(g_s))_{s \geq 0}$ is a strictly increasing sequence of positive integers bounded by $\text{index}(g)$ — contradiction. So W is a transversal. If $F \subseteq W$ is finite, then $F \subseteq I \cup Z_s$ for some s , and the member g_s is disjoint from $I \cup Z_s \supseteq F$; so F is not a transversal. $\square$

Claim (c) contradicts the Compactness Theorem of Step 4a. Hence the assumption was false and $\mathcal{C}$ is finite. $\blacksquare$

Conclusion

By Step 4b, $\mathcal{M}$ is finite; by Step 2, there exist positive integers T and L (explicitly $L = \text{lcm}(\mathcal{M})$, $T = \#(S_\infty \cap [a_1, a_1 + L))$) with $a_{n+T} = a_n + L$ for every $n \geq 1$. $\blacksquare$

Verification status of each step

- Step 1, 2: self-contained; numerically confirmed (`code/verify_newproof.py`, `code/bruteforce_check.py`).

- Step 3: 3.1–3.2 numerically confirmed for ~ 85 values of a_1 ; 3.3 (K) is a direct consequence of greedy minimality; 3.4 (K) derived from (K) + Step 2 descent — numerically confirmed in `code/check_Kprime.py`.
- Step 4a (Compactness): uses only Claims 3.1, 3.2, 3.4 + antichain; each inference checked above. The descent terminates since π drops by a factor ≥ 2 per step; pigeonhole uses finitely many prime-sets of product $\leq a_1$.
- Step 4b (peeling): Case A now uses only finite self-duality (minimal sub-transversal of a FINITE transversal is finite); Claim (c) is pure combinatorics; Compactness kills the runs-forever case. **The previously flagged gap is closed.**